

MATHEMATICAL FOUNDATIONS OF POLITICAL METHODOLOGY: PLSC 43401

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Course description

This is a first course on the theory and practice of mathematical methods in social science research. These mathematical and computer skills are needed for the quantitative and formal modeling courses offered in the political science department, and are increasingly necessary for courses in American Politics, Comparative Politics, and International Relations. We will cover mathematical techniques (linear algebra, calculus, probability) and methods of logical and statistical inference (proofs and statistics). A weekly computing lab will apply these methods, as well as introduce the R statistical computing environment.

Grading: You will be graded on weekly problem sets (worth 20% of your total grade), several programming assignments in R (also 20%), and two, noncumulative exams (each worth 30% of your grade).

Problem sets are due each Tuesday at midnight, starting October 3rd. Upload your problem sets to Canvas.

Required books: *Mathematics for Economists*, 4th Edition, by Pemberton & Rau, and *Introduction to Probability*, 2nd Edition, by Bertsekas & Tsitsiklis.

SEP 26: INTRODUCTION: ALGEBRA, SETS, FUNCTIONS, AND PROOFS

- Pemberton & Rau Ch. 1-4
- **Week's R Goal:** Intro to R, data frames, Boolean logic, indexing, loops, functions.
- **R application:** NYC Flights

SEP 28: SEQUENCES, SERIES, LIMITS & CONTINUITY

- Pemberton & Rau Ch. 5

OCT 3: THE DERIVATIVE & METHODS OF DIFFERENTIATION

- Pemberton & Rau Ch. 6, 7
- **Week's R Goal:** Vectorized code with dplyr, joins.
- **R application:** Fearon Laitin and Religion

OCT 5: OPTIMIZATION & INTEGRAL CALCULUS

- Pemberton & Rau Ch. 8, 19

OCT 10: LINEAR ALGEBRA

- Pemberton & Rau Ch. 11
- **Week's R Goal:** Graphical summary of data, histograms, bar plots, group_by.
- **R application:** Language and geography of Afghanistan's villages

OCT 12: SYSTEMS OF LINEAR EQUATIONS & DETERMINANTS

- Pemberton & Rau Ch. 12, 13.1

OCT 17: MULTIVARIATE CALCULUS

- Pemberton & Rau Ch. 14, 16
- **Week's R Goal:** tidyr
- **R application:** Police shootings

OCT 19: CONSTRAINED OPTIMIZATION

- Pemberton & Rau Ch. 16, 17

OCT 24: IN-CLASS EXAM

- **Week's R Goal:** Purrr, advanced functions on lists

OCT 26: LAWS AND PROPERTIES OF PROBABILITY

- Bertsekas-Tsitsiklis Ch. 1.1-1.2

OCT 31: CONDITIONAL PROBABILITY

- Bertsekas-Tsitsiklis Ch. 1.3-1.7
- **Week's R Goal:** xtable, stargazer, reproducible code
- **R Application:** Data vs model of World Series lengths

NOV 2: DISCRETE RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 2

NOV 7: CONTINUOUS RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 3.1-3.3
- **Week's R Goal:** stringr, and other text tools
- **R Application:** Open Data Chicago: the Wage Gap

NOV 9: SUMMARIZING RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 3.4-3.7

NOV 14: COVARIANCE AND CORRELATION

- Bertsekas-Tsitsiklis Ch. 4.1-4.3
- **Week's R Goal:** ggmap and GIS

NOV 16: TRANSFORMATIONS OF RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 4.4-4.4.6

NOV 21: INEQUALITIES AND LIMITS OF SEQUENCES OF RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 5
- **Week's R Goal:** validate
- **R Application:** China's response to the TPP - a survey experiment

NOV 28: HYPOTHESIS TESTING

- Bertsekas-Tsitsiklis Ch. 9
- Degroot and Schervish Chapter 8

NOV 30: IN-CLASS EXAM